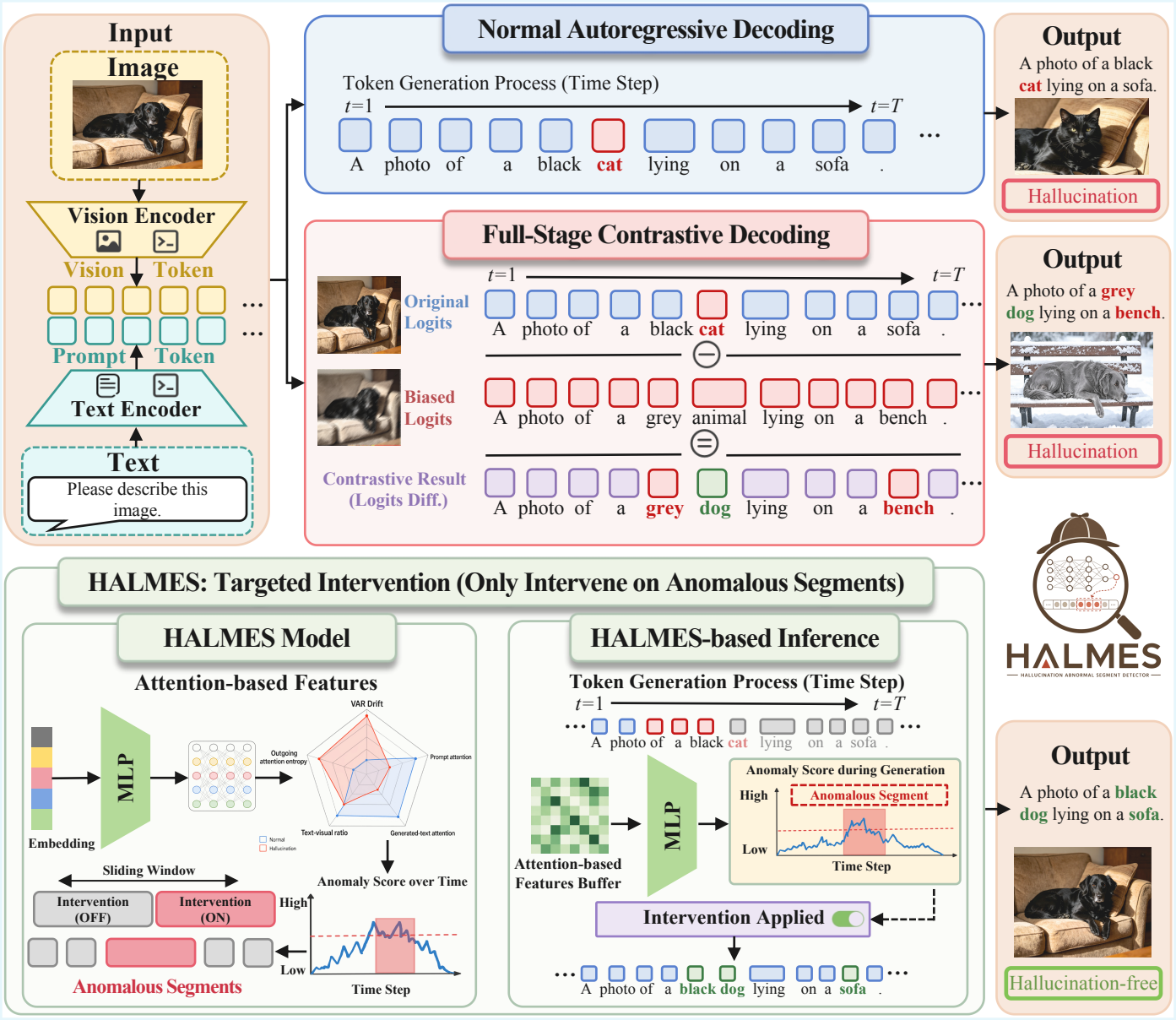

Input
Image
Vision Encoder
Vision Token
Prompt Token
Text Encoder
Text
Please describe this image.
Normal Autoregressive Decoding
Token Generation Process (Time Step)
t=1
t=T
A photo of a black cat lying on a sofa .
Output
A photo of a black cat lying on a sofa.
Hallucination
Full-Stage Contrastive Decoding
Original Logits
A photo of a black cat lying on a sofa .
Biased Logits
A photo of a grey animal lying on a bench .
Contrastive Result (Logits Diff.)
A photo of a grey dog lying on a bench .
Output
A photo of a grey dog lying on a bench.
Hallucination
HALMES
HALLUCINATION ABNORMAL SEGMENT DETECTOR
HALMES: Targeted Intervention (Only Intervene on Anomalous Segments)
HALMES Model
Attention-based Features
Embedding
MLP
VAR Drift
Prompt attention
Generated-text attention
Text-visual ratio
Outgoing attention entropy
Normal
Hallucination
Sliding Window
Intervention (OFF)
Intervention (ON)
Anomalous Segments
Anomaly Score over Time
High
Low
Time Step
HALMES-based Inference
Token Generation Process (Time Step)
t=1
t=T
A photo of a black cat lying on a sofa .
Attention-based Features Buffer
MLP
Anomaly Score during Generation
Anomalous Segment
High
Low
Time Step
Intervention Applied
A photo of a black dog lying on a sofa .
Output
A photo of a black dog lying on a sofa.
Hallucination-free